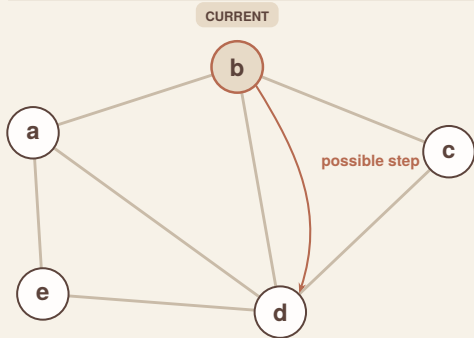


WORKED EXAMPLE · MATHEMATICAL EXPOSITION

Random walks on graphs

From local movement to graph structure and long-run behavior

Edges become transition probabilities

**DEFINITION**

Choose a neighboring vertex in proportion to edge weight.

For weighted adjacency matrix W , the transition probability is

$$P_{ij} = \frac{W_{ij}}{\sum_k W_{ik}}.$$

Every row of P sums to one. The graph specifies the available transitions.

One step is matrix multiplication

TRANSITION MATRIX

$$P = \begin{pmatrix} 0 & \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & 0 & \frac{1}{3} & \frac{1}{3} & 0 \\ 0 & \frac{1}{2} & 0 & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & 0 & \frac{1}{4} \\ \frac{1}{2} & 0 & 0 & \frac{1}{2} & 0 \end{pmatrix}.$$

At each vertex, choose one neighbor uniformly.

DISTRIBUTION UPDATE

If μ_t records the position distribution at time t , then

$$\mu_{t+1} = \mu_t P.$$

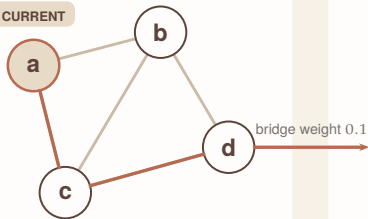
The initial row vector is concentrated at vertex b .

After one step, each of a , c , and d has probability $1/3$.

A bottleneck delays movement between two regions

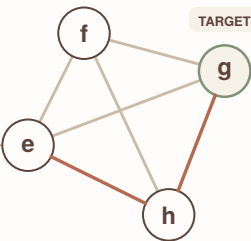
REGION A

CURRENT



REGION B

TARGET



CUT BOTTLENECK

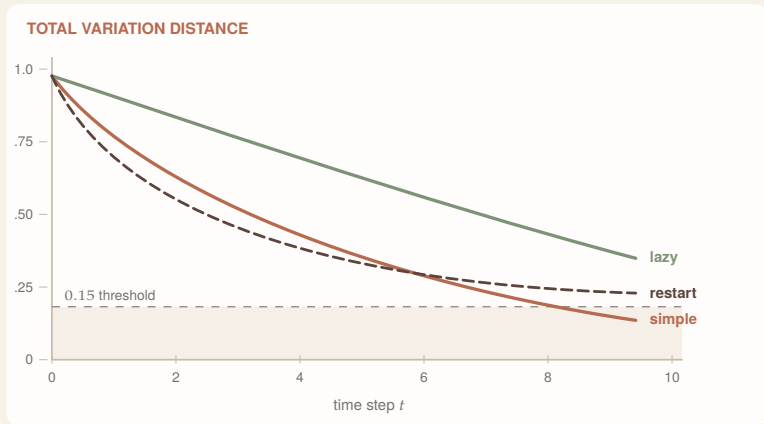
A single weak edge connects the two regions.

$$\Phi(S) = \frac{w(S, \bar{S})}{\text{vol}(S)}.$$

Small conductance means that probability leaves a region slowly.

Crossing remains possible but rare.

Walk variants produce different convergence profiles ⁰⁵



ILLUSTRATIVE READOUT

Fast early decline can end at a biased floor.

Simple walk
threshold near $t = 8$

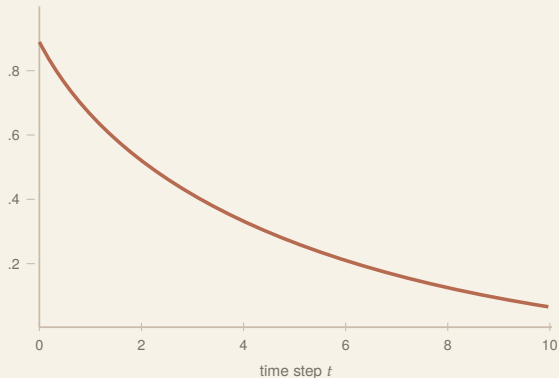
Restart walk
reaches a biased floor

Lazy walk
slower, same target

Values are schematic only.

This walk converges to one stationary distribution

DISTANCE TO STATIONARITY



STATIONARY DISTRIBUTION

$$\pi P = \pi.$$

On a finite, connected, non-bipartite graph, the walk has a unique limit.

Vertices of higher degree receive more long-run probability.

Applications, assumptions, and reporting choices

USEFUL FOR

Local diffusion

Ranking, sampling, load spreading, and simple models of movement through a network.

REQUIRES CARE

Structure matters

Direction, weights, disconnected regions, and periodicity change both movement and convergence.

REPORT EXPLICITLY

Transition rule and stopping rule

State whether the chain is weighted, lazy, restarted, or observed for a fixed number of steps.

Transition rules are simple, but graph structure determines the result.

State the rule, its consequence, and the assumptions used to interpret the result.